

LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

3 - SISO State Space Review - Continued

© Prof. Victor Solo

School of Electrical Engineering and
Telecommunications

UNSW

Sydney

Slide 1

Topics

- 1** State Space Decomposition Theorem
- 2** Polynomial Methods for coprimeness
(pole-zero cancellation) Sylvester's test (and
state space view) and Bezout test.
- 3** State Space vs Time Domain Stability

DECOMPOSITION THEOREM - Modal View

We can group the controllability coefficients β_k and observability coefficients γ_k as follows:

Label	Modes	Coefficients
$c, 0$	controllable and observable	$\beta_k \neq 0, \gamma_k \neq 0$
$c, \bar{0}$	controllable not observable	$\beta_k \neq 0, \gamma_k = 0$
$\bar{c}, 0$	observable not controllable	$\beta_k = 0, \gamma_k \neq 0$
$\bar{c}, \bar{0}$	not controllable, not observable	$\beta_k = 0, \gamma_k = 0$

This allows us to partition the state space in the same way. This is best illustrated with an example. Suppose

$$\begin{aligned}\beta^T &= (\beta_1, 0, \beta_3, \beta_4, 0) \\ \gamma^T &= (\gamma_1, \gamma_2, 0, \gamma_4, 0)\end{aligned}$$

Then reorder the states to get the following SS model

DECOMPOSITION THEOREM - Modal View - Continued

Label	x	A	β	γ
$c, 0$	x_1	$\lambda_1 \quad 0 \quad 0 \quad 0 \quad 0$	β_1	γ_1
$c, 0$	x_4	$0 \quad \lambda_4 \quad 0 \quad 0 \quad 0$	β_4	γ_4
$c, \bar{0}$	x_3	$0 \quad 0 \quad \lambda_3 \quad 0 \quad 0$	β_3	0
$\bar{c}, 0$	x_2	$0 \quad 0 \quad 0 \quad \lambda_2 \quad 0$	0	γ_2
$\bar{c}, \bar{0}$	x_5	$0 \quad 0 \quad 0 \quad 0 \quad \lambda_5$	0	0

This decomposition result is general.

Reachability & Detectability

(A, B) is stabilizable if all uncontrollable modes are stable.

(A, C) is detectable if all unobservable modes are stable.

These modes cannot be altered but they do not interfere with system operation (unless excited with large aligned initial conditions).

POLYNOMIAL DIVISION

Given polynomials

$$\begin{aligned}b(s) &= b_n + b_{n-1}s + \cdots + b_0s^n \\a(s) &= a_n + a_{n-1}s + \cdots + a_0s^n\end{aligned}$$

We calculate the transfer function

$$\begin{aligned}P(s) &= \frac{b(s)}{a(s)} \\&= \frac{b_ns^{-n} + b_{n-1}s^{-(n-1)} + \cdots + b_0}{a_ns^{-n} + a_{n-1}s^{-(n-1)} + \cdots + a_0} \\&= \sum_0^\infty g_rs^{-r}\end{aligned}$$

Multiplying across we have

$$(g_0 + g_1s^{-1} + \cdots)(a_0 + a_1s^{-1} + \cdots) = b_0 + b_1s^{-1} + \cdots$$

Equate coefficients of s^{-r} to find for any $m \geq 0$

POLYNOMIAL DIVISION Continued

$$\begin{matrix} s^0 \\ s^{-1} \\ : \\ : \\ s^{-m} \end{matrix} \begin{bmatrix} a_0 & 0 & \dots & \dots & 0 \\ a_1 & a_0 & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ a_m & a_{m-1} & \dots & \dots & a_0 \end{bmatrix} \begin{bmatrix} g_0 \\ g_1 \\ : \\ : \\ g_m \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ : \\ : \\ b_m \end{bmatrix}$$

or

$$\mathcal{A}_{-}^{(m)} g^{(m)} = b^{(m)}$$

also of course

$$\mathcal{G}_{-}^{(m)} a^{(m)} = b^{(m)}$$

Notes:

- (i) $\mathcal{A}_{-}^{(m)}$ is a lower triangular Toeplitz matrix.
- (ii) Once $m > n$, then the corresponding a, b coefficients are 0. e.g. $n = 3, m = 5$

$$\mathcal{A}_{-}^{(5)} = \begin{bmatrix} a_0 & 0 & 0 & 0 & 0 \\ a_1 & a_0 & 0 & 0 & 0 \\ a_2 & a_1 & a_0 & 0 & 0 \\ 0 & a_2 & a_1 & a_0 & 0 \\ 0 & 0 & a_2 & a_1 & a_0 \end{bmatrix}$$

Slide 7

SYLVESTER'S THEOREM

Theorem: $a(s), b(s)$ are coprime iff $\det(\mathcal{S}) \neq 0$

$$a(s) = a_n + a_{n-1}s + \dots + a_0s^n$$

$$b(s) = b_n + b_{n-1}s + \dots + b_0s^n$$

$$\mathcal{S} = \left[\begin{array}{c|c} \mathcal{A}_- & \mathcal{B}_- \\ \hline \mathcal{A}_+ & \mathcal{B}_+ \end{array} \right]$$

$$\mathcal{A}_- = \begin{bmatrix} a_0 & 0 & \dots & \dots & 0 \\ a_1 & a_0 & \dots & \dots & \dots \\ \dots & a_1 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ a_{n-1} & \dots & \dots & \dots & a_0 \end{bmatrix}$$

$$\mathcal{A}_+ = \begin{bmatrix} a_n & a_{n-1} & \dots & \dots & a_1 \\ 0 & a_n & a_{n-1} & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & a_n \end{bmatrix}$$

Similarly for $\mathcal{B}_-, \mathcal{B}_+$

SYLVESTER'S THEOREM Continued

NB. (i) $n = \max \text{ degree of } a(s), b(s)$

(ii) $\mathcal{A}_-$ is lower triangular Toeplitz with first column $(a_0 \dots a_{n-1})^T$

$\mathcal{A}_+$ is upper triangular Toeplitz with first row $(a_n \dots a_1)$.

Proof

(i) If $a(s), b(s)$ have a common root λ_0 then $a(s) = (s - \lambda_0)\bar{a}(s)$, $b(s) = (s - \lambda_0)\bar{b}(s)$ and $\bar{a}(s), \bar{b}(s)$ have degree $n - 1$.

Eliminate $s - \lambda_0$ to find

$$a(s)\bar{b}(s) - \bar{a}(s)b(s) = 0$$

Equating both sides gives $2n$ equations
e.g. for $n = 3$ we get $2n = 6$ equations

$$\mathcal{A}_-^{(6)}\bar{b} = \mathcal{B}_-^{(6)}\bar{a}$$

SYLVESTER'S THEOREM Continued II

e.g. the first term is

$$\mathcal{A}_{-}^{(6)}\bar{b} = \begin{bmatrix} a_0 & 0 & 0 & \cdot & 0 & 0 & 0 \\ a_1 & a_0 & 0 & \cdot & 0 & 0 & 0 \\ a_2 & a_1 & a_0 & \cdot & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ a_3 & a_2 & a_1 & \cdot & a_0 & 0 & 0 \\ 0 & a_3 & a_2 & \cdot & a_1 & a_0 & 0 \\ 0 & 0 & a_3 & \cdot & a_2 & a_1 & a_0 \end{bmatrix} \begin{bmatrix} \bar{b}_0 \\ \bar{b}_1 \\ \bar{b}_2 \\ \cdot \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

This has a nice partitioned structure (and the other term similarly) as follows

$$\begin{bmatrix} \mathcal{A}_{-}^{(3)} & 0 \\ \mathcal{A}_{+}^{(3)} & \mathcal{A}_{-}^{(3)} \end{bmatrix} \begin{bmatrix} \bar{b} \\ 0 \end{bmatrix} - \begin{bmatrix} \mathcal{B}_{-} & 0 \\ \mathcal{B}_{+} & \mathcal{B}_{-} \end{bmatrix} \begin{bmatrix} \bar{a} \\ 0 \end{bmatrix} = 0$$

Now write out the partitioned equations

$$\mathcal{A}_{-}\bar{b} - \mathcal{B}_{-}\bar{a} = 0$$

$$\mathcal{A}_{+}\bar{b} - \mathcal{B}_{+}\bar{a} = 0$$

SYLVESTER'S THEOREM Continued

or

$$\begin{bmatrix} \mathcal{A}_- & \mathcal{B}_- \\ \mathcal{A}_+ & \mathcal{B}_+ \end{bmatrix} \begin{bmatrix} \bar{b} \\ -\bar{a} \end{bmatrix} = 0$$

$$\Rightarrow \mathcal{S}\theta = 0, \theta^T = (\bar{b}^T, -\bar{a}^T)$$

Now this equation has a solution iff $\det(\mathcal{S}) = 0$

(ii) If $\det(\mathcal{S}) = 0$, then the equation $\mathcal{S}\theta = 0$ has a solution. Now run the above argument backwards to find

$$a(s)\bar{b}(s) = \bar{a}(s)b(s)$$

If λ_0 is a root of $a(s)$, then

$$0 = a(\lambda_0)\bar{b}(\lambda_0) = \bar{a}(\lambda_0)b(\lambda_0)$$

So either $\bar{a}(\lambda_0) = 0$ or $b(\lambda_0) = 0$ or both. But $\bar{a}(s)$ has only $n - 1$ roots whereas both $b(s), a(s)$ have n roots. So for at least one of these roots we must have $b(\lambda_0) = 0$. Hence as required a common factor exists.

SYLVESTER'S TEST - Example

Test $a(s), b(s)$ for coprimeness with

$$a(s) = s^3 - 8s^2 + 19s - 12, \quad b(s) = s^3 - 7s + 6$$

We form the Sylvester matrix

$$\mathcal{S} = \begin{pmatrix} 1 & 0 & 0 & \cdot & 1 & 0 & 0 \\ -8 & 1 & 0 & \cdot & 0 & 1 & 0 \\ -19 & -8 & 1 & \cdot & -7 & 0 & 1 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -12 & 19 & -8 & \cdot & 6 & -7 & 0 \\ 0 & -12 & 19 & \cdot & 0 & 6 & -7 \\ 0 & 0 & -12 & \cdot & 0 & 0 & 6 \end{pmatrix}$$

Matlab SVD delivers following singular values

36.2, 23.3, 10.2, 0.13, 0.00

showing $\mathcal{S}$ is singular $\Rightarrow$ common factor in $a(s), b(s)$.

NB. Common root is $s = 1$ since

$$a(s) = (s-3)(s-1)(s-4), \quad b(s) = (s+3)(s-1)(s-2)$$

POLE-ZERO CANCELLATION - State Space View vs Polynomial View

For the state space model A, b, c the transfer function is $c^T(sI - A)^{-1}b = \frac{b(s)}{a(s)}$, say. We know that there is pole-zero cancellation. i.e. $b(s), a(s)$ have a common factor iff $\mathcal{C}(A, b)$ or $\mathcal{O}(c, A)$ is singular. So what is the connexion to the Sylvester test based on polynomials?

Kailath p.102 shows for the controller form

$$\mathcal{O}_c = \mathcal{O}(c_c, A_c) = (\mathcal{B}_+ - \mathcal{B}_- \mathcal{A}_-^{-1} \mathcal{A}_+) \tilde{I}$$

Where $\tilde{I}$ has 1's down reverse diagonal and zeros elsewhere. But by determinant partitioning lemma

$$|\mathcal{S}| = |\mathcal{A}_-| |\mathcal{B}_+ - \mathcal{A}_+ \mathcal{A}_-^{-1} \mathcal{B}_-|$$

But lower triangular matrices commute so

$$= |\mathcal{A}_-| |\mathcal{B}_+ - \mathcal{B}_- \mathcal{A}_-^{-1} \mathcal{A}_+| = |\mathcal{A}_-| |\mathcal{O}_c|$$

$$\text{So } |\mathcal{S}| = 0 \text{ iff } |\mathcal{O}_c| = 0$$

So the tests do the same thing.

Slide 13

BEZOUT LEMMA

$a(s)$ of degree n , $b(s)$ of degree m are coprime iff there exist polynomials $\tilde{x}(s)$ of degree $\leq m - 1$, $\tilde{y}(s)$ degree $\leq n - 1$ (which are then unique) such that

$$\tilde{x}(s)a(s) + \tilde{y}(s)b(s) = 1 \quad (0.1)$$

Proof. Take $m = n$ for simplicity. If there exist $\tilde{x}(s), \tilde{y}(s)$ equate coefficients to find the $2n$ equations (with $\tilde{x}, \tilde{y}$ being n vectors)

$$\begin{pmatrix} \mathcal{A}_- & 0 \\ \mathcal{A}_+ & \overline{\mathcal{A}}_- \end{pmatrix} \begin{pmatrix} \tilde{x} \\ 0 \end{pmatrix} + \begin{pmatrix} \mathcal{B}_- & 0 \\ \mathcal{B}_+ & \overline{\mathcal{B}}_- \end{pmatrix} \begin{pmatrix} \tilde{y} \\ 0 \end{pmatrix} = \begin{pmatrix} e_1 \\ 0 \end{pmatrix}$$

Collect these equations together as

$$\begin{aligned} \mathcal{A}_- \tilde{x} + \mathcal{B}_- \tilde{y} &= e_1 \\ \mathcal{A}_+ \tilde{x} + \mathcal{B}_+ \tilde{y} &= 0 \end{aligned}$$

or

$$\begin{pmatrix} \mathcal{A}_- & \mathcal{B}_- \\ \mathcal{A}_+ & \mathcal{B}_+ \end{pmatrix} \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix} = \begin{pmatrix} e_1 \\ 0 \end{pmatrix} \quad (0.2)$$

BEZOUT LEMMA Continued

This equation has a unique a solution iff $\det(\mathcal{S}) \neq 0$.

If $a(s), b(s)$ are coprime then $\det(\mathcal{S}) \neq 0$ and so (0.2) has a solution $\tilde{x}, \tilde{y}$ and running the discussion in reverse constructs $\tilde{x}(s), \tilde{y}(s)$ satisfying (0.1).

If $a(s), b(s)$ are not coprime let λ_0 be the common root and we can factor

$$a(s) = (s - \lambda_0)\bar{a}(s), b(s) = (s - \lambda_0)\bar{b}(s)$$

So (0.1) becomes

$$(s - \lambda_0) [\tilde{x}(s)\tilde{a}(s) + \tilde{y}(s)\bar{b}(s)] = 1$$

and setting $s = \lambda_0$ gives $0 = 1$ a contradiction. So if (0.1) holds $a(s), b(s)$ must be coprime.

Diophantine Equations

Suppose we have polynomials $a(s)$ of degree n and $b(s)$ of degree $m \leq n$ and $c(s)$ of degree r . In control design the following diophantine equation occurs frequently. Solve $a(s)f(s) + b(s)g(s) = c(s)$ for polynomials $f(s), g(s)$.

Result I. Suppose $a(s), b(s)$ are coprime and let $\tilde{x}(s), \tilde{y}(s)$ be the unique solution to $a(s)\tilde{x}(s) + b(s)\tilde{y}(s) = 1$ then the general solution to the diophantine equation is,

$$f(s) = c(s)\tilde{x}(s) - b(s)p(s),$$

$$g(s) = c(s)\tilde{y}(s) + a(s)p(s)$$

where $p(s)$ is an arbitrary polynomial.

Result II. If we add the requirement that $\deg c = m + n - 1$ and $\deg(f) < m, \deg(g) < n$ then we can produce a unique solution by polynomial division. We find by division a polynomial $l(s)$ with,

$$f(s) = l(s)b(s) + f_0(s) \& \deg(f_0) < m. \text{ Then set}$$

$$g_0(s) = g(s) + l(s)a(s); \text{ the unique solution is}$$

$$\text{then } f_0(s), g_0(s) \text{ and } \deg(g_0) < n.$$

Diophantine I Proof

Dropping the s argument we have

$$af + bg = ac\tilde{x} - abp + bc\tilde{y} + bap = c(a\tilde{x} + b\tilde{y}) = c.$$

On the other hand if $af + bg = c$ then set

$$\alpha(s) = f - c\tilde{x}, \beta(s) = g - c\tilde{y}. \text{ Then}$$

$$a\alpha + b\beta = af - ac\tilde{x} + bg - bc\tilde{y} = c - c(a\tilde{x} + b\tilde{y}) =$$

$$c - c = 0 \Rightarrow a\alpha = -b\beta \Rightarrow \frac{\alpha}{\beta} = -\frac{b}{a}. \text{ But since } a, b$$

are coprime we must have for some polynomial

$$p(s) \text{ that } \alpha = -bp, \beta = ap \text{ and the result follows.}$$

Diophantine II Proof

By division we can find

$f(s) = l(s)b(s) + f_0(s)$ & $\deg(f_0) < m$. Then set

$g_0(s) = g(s) + l(s)a(s)$. Then we have

$$a(s)f_0(s) + b(s)g_0(s) = a(s)f(s) - a(s)b(s)l(s) + b(s)g(s) + b(s)l(s)a(s) = a(s)f(s) + b(s)g(s) = c(s).$$

Now suppose $f_1(s), g_1(s)$ is another solution satisfying the degree constraints.

Then $af_0 + bg_0 = c, af_1 + bg_1 = c$ and subtracting gives $a(f_0 - f_1) + b(g_0 - g_1) = 0 \Rightarrow \frac{f_0 - f_1}{g_0 - g_1} = -\frac{b}{a}$.

Since a, b are coprime this means there is a

polynomial $l(s)$ with $f_0 - f_1 = -lb, g_0 - g_1 = la$.

For the first equation the LHS has degree $< m$

but the RHS has degree $\geq m$. This is a

contradiction unless $l(s) = 0 \Rightarrow f_0 = f_1$. But then

in the second equation $g_0 = g_1$. However in

$af_0 + bg_0 = c$ the RHS has degree $m + n - 1$ and

since b has degree m we must have

$\deg(g_0) \leq n - 1$ as required.

LTI - BIBO STABILITY

- Consider the SS model

$$\dot{x} = Ax + bu, y = c^T x$$

with transfer function

$$P(s) = c^T (sI - A)^{-1} b$$

By partial fraction expansion or SS modal decomposition (λ_k are eigenvalues of A)

$$\begin{aligned} P(s) &= \sum_1^n \frac{\gamma_k \beta_k}{s - \lambda_k} \leftrightarrow \sum_1^n \gamma_k \beta_k e^{\lambda_k t} \\ \beta_k &= q_k^T b, \gamma_k = p_k^T c \end{aligned}$$

BIBO stability follows iff $Re(\lambda_k) < 0$.

We now give a SS version of this classic stability result.

LTI - BIBO STABILITY Continued

•Lyapunov Stability

A has all eigenvalues with negative real part, i.e.

A is a stability matrix iff for a given positive definite matrix Q there is a positive definite matrix P such that

$$AP + PA^T = -Q$$

Proof. If A is a stability matrix, define

$$P = \int_0^{\infty} e^{At} Q e^{A^T t} dt$$

P is positive definite since for any vector a and $T > 0$

$$\begin{aligned} a^T P a &\geq \int_0^T a^T e^{At} Q e^{A^T t} a dt \\ &\geq \int_0^T 0 dt = 0 \end{aligned}$$

LTI - BIBO STABILITY Continued

Next consider

$$\begin{aligned} AP + PA^T &= \int_0^\infty Ae^{At}Qe^{A^T t}dt + PA^T \\ &= \int_0^\infty \left(\frac{d}{dt}e^{At} \right) Qe^{A^T t}dt + PA^T \\ &= e^{At}Qe^{A^T t} \Big|_0^\infty \\ &\quad - \int_0^\infty e^{At}Qe^{A^T t}A^T dt + PA^T \\ &= -Q - PA^T + PA^T = -Q \end{aligned}$$

as required.

Conversely, if $PA + A^T P = -Q$. Let q be an eigenvector of A^T with eigenvalue λ then

$$\begin{aligned} q^H APq + q^H PA^T q &= -q^H Qq \\ &= \lambda^H q^H Pq + q^H Pq\lambda \\ &= 2\operatorname{Re}(\lambda)q^H Pq \\ \Rightarrow 2\operatorname{Re}(\lambda) &= -\frac{q^H Qq}{q^H Pq} < 0 \end{aligned}$$

since both Q, P are positive definite.

Slide 21